

HW/Lab 3 Sol: Multipath Channel and OFDM

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Theory Problems

1. OFDM Theory

An OFDM system operates at a carrier frequency of 2.0 GHz with a total bandwidth of 5 MHz. The channel has a maximum delay spread of $\tau_{\max} = 4 \mu s$. Design the OFDM numerology such that the cyclic prefix overhead is 10% (i.e., 90% efficiency). Assume that the entire 5 MHz bandwidth is used for subcarriers, without allocating guard bands.

- a) Determine the useful OFDM symbol duration T_{ofdm} and cyclic prefix duration T_{cp} .

Solution:

Since delay spread is $4 \mu s$, the CP length should be at least $4 \mu s$. To maximize efficiency, choose $T_{cp} = 4 \mu s$. For getting 90% efficiency with CP, the symbol time $T_{ofdm} = 36 \mu s$.
(The packet time becomes $40 \mu s$, efficiency $\frac{T_{ofdm}}{T_{ofdm} + T_{cp}} = 90\%$).

- b) Find the subcarrier spacing Δf and calculate the number of subcarriers N required.

Solution:

The subcarrier spacing is $\frac{1}{T_{ofdm}} = \frac{1}{36 \mu s} = 27.78 \text{ kHz}$. Since bandwidth is 5 MHz, we have number of subcarriers to be $5 \text{ MHz} * 36 \mu s$, which amount to 180 subcarriers.

- c) Suggest a practical FFT size (power of two) that would be chosen in an implementation.

Solution:

If we need to select a power of 2, we would go ahead with 256 subcarriers.

- d) (ECSE 6560 only) Now repeat the design for the case when the OFDM system supports mobility up to 360 km/h. Compute the maximum Doppler shift at $f_c = 2.0 \text{ GHz}$ and estimate the corresponding channel coherence time.

Solution:

Now, we have mobile speed as 360 kmph, which gives maximum Doppler shift

$f_d = \frac{v}{\lambda} = \frac{v}{c} f_c = 666.67 \text{ Hz}$, and coherence time $\frac{1}{10 f_d} = \frac{\lambda}{10 v} = 150 \mu s$. (Other rules for coherence time are acceptable.)

- e) (ECSE 6560 only) Based on the coherence time, propose a reference signal (pilot) design that maintains 90% efficiency, and illustrate the pilot arrangement with a labeled time-frequency grid figure (showing OFDM symbols on the time axis and subcarriers on the frequency axis).

Solution:

If we follow the preamble estimation paradigm, we would have to resend the the preamble after each 4 symbols of $40 \mu s$ time. Hence, the role of pilots in estimating and track the channel is more significant.

Alongwith this, the moving user would introduce doppler shift, which would manifest as SFO while equalizing. Here, to mitigate this, pilots need to be stuffed in with more density, so as to estimate the linear phase shift before the 2π jumps. Hence, the pilot placement has to be done in a manner which allows estimation of multiple 2π jumps within the symbol period. For example, say within $40\mu s$, the SFO phase incurs three 2π jumps. Then, we would be placing 12 pilots in the symbol, equispaced. The rationale behind this is that we can estimate one linear phase changes with 4 pilots, and their are 3 linear phase changes to be estimated.

Also, due to the pilots, the efficiency calculation has to be redone, for example of 12 pilots case, we need to design the system with say $\sim 95\%$ efficiency, so that the reduction in efficiency by pilots is accounted for. To get 95% efficiency, the symbol time would be $76 \mu s$. With 12 pilots among 180 subcarriers, the useful time further drops to $168/180$, which gives efficiency of 89%. Figure 1 shows packet structure with CP, data.

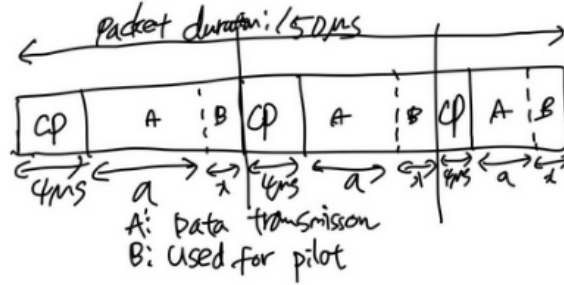


Figure 1: Frame structure

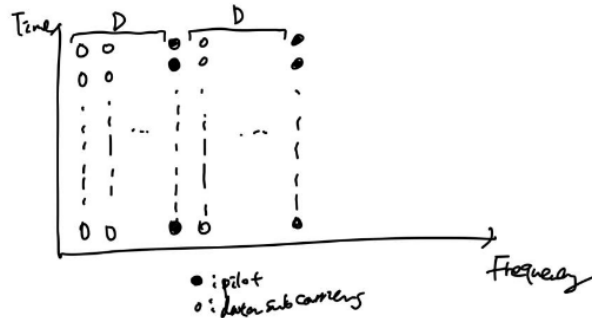


Figure 2: Resource Block

2. OFDM Orthogonality

In OFDM, data symbols are mapped onto closely spaced subcarriers that remain mutually orthogonal over the useful symbol interval. Consider the time-limited complex exponentials

$$b_k(t) = \begin{cases} e^{j2\pi f_k t}, & 0 \leq t \leq T_s, \\ 0, & \text{otherwise,} \end{cases} \quad k \in \{0, 1, \dots, N-1\},$$

and the OFDM signal

$$S(t) = \frac{1}{N} \sum_{k=0}^{N-1} d_k b_k(t),$$

where d_k are data symbols (e.g., BPSK/QPSK/QAM). Define the inner product

$$\langle b_i, b_j \rangle = \frac{1}{T_s} \int_0^{T_s} b_i(t) b_j^*(t) dt.$$

Answer the following:

- a) For which condition on i, j is $\langle b_i, b_j \rangle = 1$?

Solution:

$$\langle b_i, b_j \rangle = \frac{1}{T_s} \int_0^{T_s} e^{j2\pi(f_i - f_j)t} dt = \begin{cases} 1, & f_i = f_j, \\ \frac{e^{j2\pi(f_i - f_j)T_s} - 1}{j2\pi(f_i - f_j)T_s}, & f_i \neq f_j. \end{cases}$$

$$\langle b_i, b_j \rangle = 1 \iff \boxed{i = j \text{ (i.e., } f_i = f_j)}.$$

- b) Let $\Delta f = f_i - f_j$. For which condition on f_i, f_j is $\langle b_i, b_j \rangle = 0$?

Solution:

For $f_i \neq f_j$, the inner product is zero when $e^{j2\pi(f_i - f_j)T_s} = 1$:

$$\langle b_i, b_j \rangle = 0 \iff (f_i - f_j)T_s \in \mathbb{Z} \setminus \{0\} \iff \boxed{f_i - f_j = \frac{m}{T_s}, m \in \mathbb{Z} \setminus \{0\}}.$$

- c) (ECSE 6560 only) What is the *minimum* subcarrier spacing $\Delta f_{\min}$ that ensures orthogonality for a given T_s ?

Solution:

The smallest nonzero spacing that satisfies orthogonality is

$$\Delta f_{\min} = \boxed{\frac{1}{T_s}}.$$

3. OFDM Example with 5G New Radio

A 5G NR base-station is operating with the following specifications:

- **Time-domain structure:** A radio frame has a duration of 10 ms and is divided into 10 subframes, each of 1 ms. Each subframe contains 2^μ slots, where μ is the numerology index. For subcarrier spacing $\Delta f = 30$ kHz, we have $\mu = 1$, so each subframe contains 2 slots. Each slot consists of 14 OFDM symbols.

- **Frequency-domain structure:** The system uses 1272 subcarriers, each spaced 30 kHz apart. Assume all subcarriers are active for data transmission.
 - **Modulation:** Each subcarrier is modulated using QPSK.
 - **Simplifying assumption:** Ignore control channels, reference signals, or guard bands (i.e., assume 100% of the grid is used for data).
- a) Draw a 2D time-frequency resource grid (time on the x -axis, subcarriers on the y -axis). Clearly label the hierarchy: frames, subframes, slots, OFDM symbols, and subcarriers.
Solution:

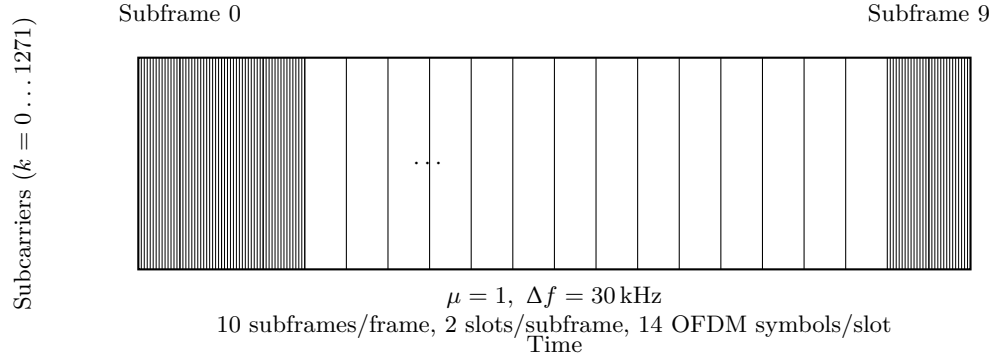


Figure 3: NR resource grid for one 10 ms frame at $\Delta f = 30 \text{ kHz}$ ($\mu = 1$): 10 subframes $\times$ 2 slots/subframe $\times$ 14 symbols/slot. All 1272 subcarriers are assumed active for data.

- b) Calculate the maximum theoretical throughput (in bits/s) under the above assumptions. Proceed as follows:

- i. Determine the number of bits carried by one subcarrier with QPSK modulation.

Solution:

QPSK carries

$$\boxed{2 \text{ bits/subcarrier}} .$$

- ii. Calculate the number of bits per OFDM symbol (using all 1272 subcarriers).

Solution:

Bits per OFDM symbol (all 1272 subcarriers active):

$$1272 \times 2 = \boxed{2544 \text{ bits/symbol}} .$$

- iii. Find the total number of OFDM symbols transmitted per second.

Solution:

OFDM symbols per second:

$$10 \text{ subframes/frame} \times 2 \text{ slots/subframe} \times 14 \text{ symbols/slot} \times 100 \text{ frames/s} = \boxed{28,000 \text{ symbols/s}} .$$

iv. Compute the final throughput in bits/s.

Solution:

$$2544 \text{ bits/symbol} \times 28,000 \text{ symbols/s} = \boxed{71,232,000 \text{ bits/s} = 71.232 \text{ Mbps}} .$$

Matlab Simulations: Pilot-Only OFDM Channel Estimation

a) Baseline

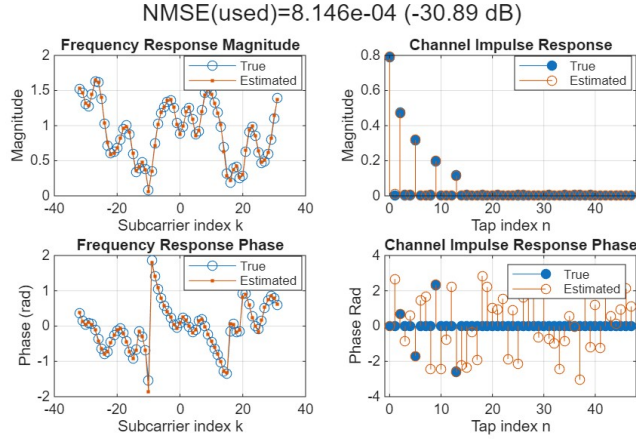


Figure 4: **Observation:** Because of CP longer than all tap delays and perfect alignment, the used NMSE is small at 30dB. The mismatch of estimated phases in the time domain is because the corresponding magnitudes are close to zero.

b) DC subcarrier nulling

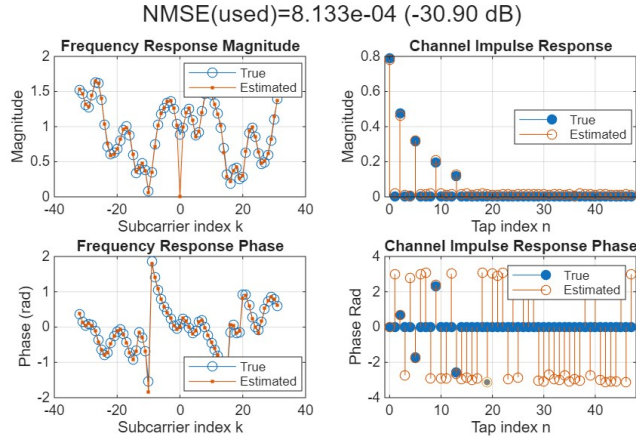


Figure 5: **Observation:** (i) DC bin is nulled in practice, because there could be DC leakage in the receiver hardware. (ii) When the DC bin is nulled, the estimated magnitude at zero frequency is 0, while in the enabled case, the estimated magnitude at zero frequency is close to its neighbors. No significant change in the estimated phase. (iii) No significant impact on NMSE, which is computed only on the used tones.

c) Guard bands

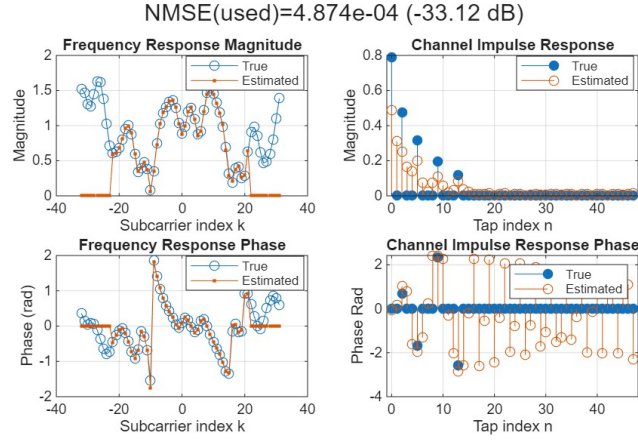


Figure 6: **Observation:** The outer 10 bins on both sides of the spectrum are nulled. The guard bands avoid strong leakage from the sinc-like peaks of the outer bins. No significant change in NMSE.

d) Noise stress test

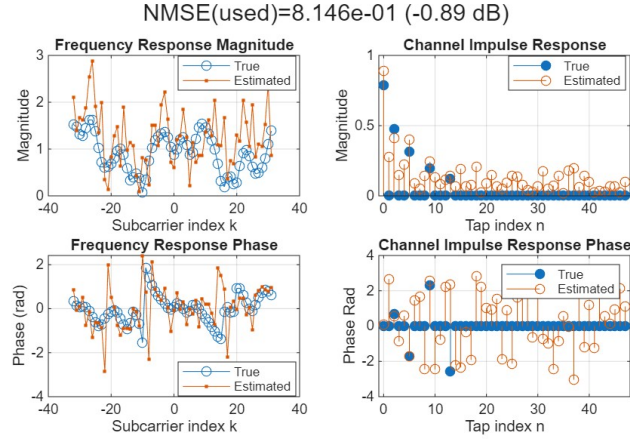


Figure 7: **Observation:** Both estimated magnitudes and phases are significantly influenced. The estimated magnitude fluctuates around the true channel in the time domain. The random noise is considered part of the fixed channel by the estimator. The NMSE is much larger than the Baseline.

e) Too-short CP (ISI)

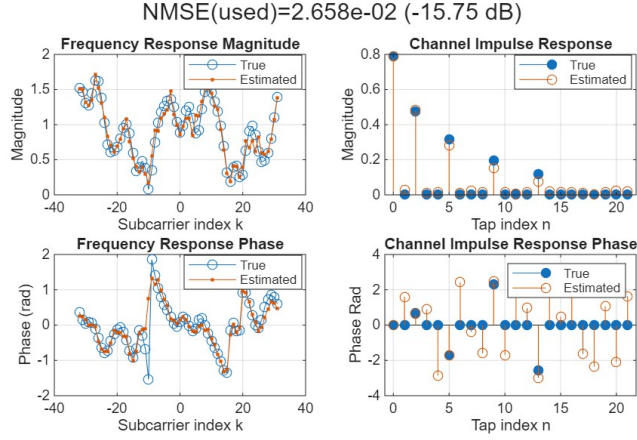


Figure 8: **Observation:** When the CP is not long enough, there are inter-symbol overlaps at the receiver, causing ISI and breaking the per-tone model equation. The overlaps also break the orthogonality between sub-channels. As a result, there are distortions in the frequency and time domains. The NMSE significantly increases as printed.

f) Timing misalignment within CP

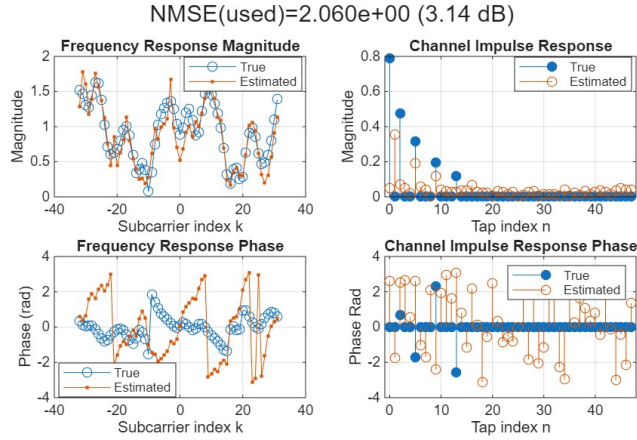


Figure 9: **Observation:** NMSE increases due to the misalignment.

MATLAB code as below.

```

1
2 % Pilot-only OFDM channel estimation demo
3 % 1) Place pilots on subcarriers (frequency domain)
4 % 2) IFFT to time, add CP
5 % 3) Pass through multipath channel (linear convolution) + AWGN
6 % 4) Remove CP, FFT
7 % 5) Per-subcarrier channel estimation (Hhat = Y/X)
8 % 6) Plot true vs. estimated channel in frequency and time domains
9 % 7) Show

```

```

10
11
12 clearvars
13 rng(123);
14
15 %% Parameters
16 N      = 64;           % FFT size
17 CP      = 24;          % CP length
18 SNRdB   = 30;          % AWGN SNR (dB); set inf for no noise
19 useDC    = true;       % set true to use DC subcarrier (usually false
    )
20 useGuards = false;     % set false to remove Guard bands
21 guard = 10;           % guard band #subcarriers one of the two edges
22
23 % Multipath channel (discrete-time tapped delay line)
24 % delays in samples, complex gains
25 tap_delays = [0 2 5 9 13];
26 tap_gains   = [1.0 0.6 0.4 0.25 0.15] .* exp(1j*2*pi*[0.00 0.11 -0.27
    0.37 -0.41]);
27 tap_gains   = tap_gains / sqrt(sum(abs(tap_gains).^2)); % normalize
    power to 1
28
29 %% Pilot allocation
30 % Fill all usable subcarriers with BPSK pilots
31
32 pilotF = randi([0 1], N, 1) * 2 - 1; % values in {+1, -1} pilot value
    on each subcarrier
33 if ~useDC
34     pilotF(N/2+1) = 0; % DC bin (center) set to zero
35 end
36 if useGuards
37     pilotF(1:guard) = 0; % here we simply null a few bins at
    edges
38     pilotF(end-guard+1:end) = 0;
39 end
40
41
42 %% TX: IFFT, CP append
43 % NOTE: We'll treat 'pilotF' as an FFT-shifted vector for visualization
    .
44 % For actual IFFT, we need unshifted frequency ordering.
45 pilotF_unshift = ifftshift(pilotF); % put DC at index 1 for ifft
    ()
46 x_noCP = ifft(pilotF_unshift, N); % time-domain symbol (length
    N)
47 x_tx = [x_noCP(end-CP+1:end); x_noCP]; % prepend CP (length N+CP)
48
49 %% Channel

```

```

50 % Build discrete LTI channel impulse response h (length Lh = max delay
    +1)
51 Lh = max(tap_delays) + 1;
52 h = zeros(Lh,1);
53 h(tap_delays + 1) = tap_gains;
54
55 % The "true" frequency response for circular convolution of length N
56 % is the N-point DFT of h zero-padded to length N, viewed with fftshift
    .
57 H_true_unshift = fft([h; zeros(N - numel(h), 1)], N);
58 H_true = fftshift(H_true_unshift);
59
60 freq_axis = -N/2:N/2-1; % shifted freq axis
61
62 % For comparison, we plot first, say, 2*CP taps
63 plot_taps = min(N, max(2*CP, Lh + 8)); % a reasonable window
64
65 %% Plot ground truth True channel
66 figure(1); clf;
67 subplot(2,2,1);
68 plot(freq_axis, abs(H_true), 'o-', 'DisplayName','True'); hold on;
69 xlabel('Subcarrier index k'); ylabel('Magnitude');
70 title('Frequency Response Magnitude');
71 legend('Location','northeast');
72
73 subplot(2,2,3);
74 plot(freq_axis, angle(H_true), 'o-', 'DisplayName','True'); hold on;
75 xlabel('Subcarrier index k'); ylabel('Phase (rad)');
76 title('Frequency Response Phase');
77 legend('Location','northeast');
78
79 % figure(2); clf;
80 subplot(2,2,2);
81 stem(0:plot_taps-1, abs([h; zeros(plot_taps-numel(h),1)]), 'filled', '
    DisplayName','True'); hold on;
82 xlabel('Tap index n'); ylabel('Magnitude');
83 title('Channel Impulse Response');
84 legend('Location','northeast');
85
86 subplot(2,2,4);
87 stem(0:plot_taps-1, angle([h; zeros(plot_taps-numel(h),1)]), 'filled',
    'DisplayName','True'); hold on;
88 xlabel('Tap index n'); ylabel('Phase Rad');
89 title('Channel Impulse Response');
90 legend('Location','northeast');
91
92
93 % All blocks below are TODO

```

```

94
95 %% Apply channel
96
97 % Linear convolution (physical multipath)
98 y_lin = conv(x_tx, h); % length N+CP+Lh-1
99
100 % Add AWGN
101 y_rx = awgn(y_lin, SNRdB, 'measured');
102
103 %% RX: Remove CP, take FFT
104
105 % For a single symbol, emulate a perfectly timed receiver that drops CP
106 % We take the window aligned to the symbol after channel: start after
    CP
107 % NOTE: Due to linear convolution, the last (Lh-1) samples are
    transient.
108 % We'll take the N samples starting from CP; i.e.; drop the CP and
109 % transient samples
110 %start_idx = CP +1; % skip CP and channel ramp-up
111 start_idx = CP +1 +4; % add misalignment
112 stop_idx = start_idx + N - 1;
113
114 y_rx_sym = y_rx(start_idx:stop_idx); % length N
115
116
117 Y_unshift = fft(y_rx_sym, N); % N-point FFT
118 Y = fftshift(Y_unshift); % for pretty plotting (DC centered)
119
120 % Known pilot in frequency (used bins)
121 X = pilotF;
122
123 %% OFDM Channel estimate
124 Hhat = complex(zeros(N,1));
125 useMask = (pilotF ~= 0); % a mask of "used" subcarriers (non-zero
    pilots)
126 Hhat(useMask) = Y(useMask) ./ X(useMask); % LS on used tones
127 Hhat(~useMask) = 0; % leave zeros on nulled
    tones
128
129 % Estimated "circular" impulse response via IDFT of Hhat
130 hhat_circ_unshift = ifft(ifftshift(Hhat), N); % circular (wrapped)
    channel
131
132
133 %% Plots
134 figure(3); clf;
135 subplot(2,2,1);
136 plot(freq_axis, abs(H_true), 'o-', 'DisplayName', 'True'); hold on;

```

```

137 plot(freq_axis, abs(Hhat), '.-', 'DisplayName','Estimated'); grid on;
138 xlabel('Subcarrier index k'); ylabel('Magnitude');
139 title('Frequency Response Magnitude');
140 legend('Location','best');
141
142 subplot(2,2,3);
143 plot(freq_axis, angle(H_true), 'o-', 'DisplayName','True'); hold on;
144 plot(freq_axis, angle(Hhat), '.-', 'DisplayName','Estimated'); grid on;
145 xlabel('Subcarrier index k'); ylabel('Phase (rad)');
146 title('Frequency Response Phase');
147 legend('Location','best');
148
149
150 subplot(2,2,2);
151 stem(0:plot_taps-1, abs([h; zeros(plot_taps-numel(h),1)]), 'filled', '
    DisplayName','True'); hold on;
152 stem(0:plot_taps-1, abs(hhat_circ_unshift(1:plot_taps)), 'DisplayName',
    'Estimated'); grid on;
153 xlabel('Tap index n'); ylabel('Magnitude');
154 title('Channel Impulse Response');
155 legend('Location','best');
156
157 subplot(2,2,4);
158 stem(0:plot_taps-1, angle([h; zeros(plot_taps-numel(h),1)]), 'filled',
    'DisplayName','True'); hold on;
159 stem(0:plot_taps-1, angle([hhat_circ_unshift(1:plot_taps)]), '
    DisplayName','Estimated'); grid on;
160
161 xlabel('Tap index n'); ylabel('Phase Rad');
162 title('Channel Impulse Response Phase');
163 legend('Location','northeast');
164
165 % NMSE over used tones
166 nmse = sum(abs(Hhat(useMask)-H_true(useMask)).^2)/sum(abs(H_true(
    useMask)).^2);
167 nmse_dB = 10*log10(nmse);
168
169 sgtitle(sprintf('NMSE(used)=%.3e (%.2f dB)',nmse,nmse_dB))
170
171
172 % figure(4);clf;
173 % stem(0:plot_taps-1, abs([h; zeros(plot_taps-numel(h),1)]), 'filled',
    'DisplayName','True'); hold on;
174 % stem(0:plot_taps-1, abs(hhat_circ_unshift(1:plot_taps)), 'DisplayName
    ','Estimated'); grid on;
175 % xlabel('Tap index n'); ylabel('Magnitude');
176 % title('Channel Impulse Response');
177 % legend('Location','best');

```